

AirClip

Tagline: Seamless, real-time clipboard synchronization across all your devices and collaboration with your team.

Overview

AirClip is a modern, web-based platform designed to eliminate the friction of transferring text, links, and small snippets of information between your devices. It uses a real-time WebSocket connection to instantly sync your clipboard in a private, secure room.

Beyond personal use, AirClip extends its real-time capabilities to collaborative environments, allowing multiple users to join custom rooms and share a live, synchronized clipboard, making it an ideal tool for pair programming, brainstorming sessions, and team-based workflows.

Key Features

1. Real-Time Clipboard Sync (Core)

- **Multi-Device Synchronization:** Instantly syncs clipboard content across multiple logged-in devices (e.g., laptop, desktop, mobile) by joining a private, user-specific room.
- **Automatic Copy:** When a new item is received from another one of your devices, it is automatically copied to the local clipboard, providing a seamless workflow.
- **Manual Copy:** A "copy" icon is available next to each item in the history for easy manual copying.

2. Collaboration Spaces (Rooms)

- **Custom Room Creation:** Users can create or join shared rooms by simply entering a room name.
- **Team Collaboration:** All users in a custom room share the same real-time clipboard, allowing for instant sharing of code snippets, links, and notes.

- **Room-Specific History:** The clipboard history is unique to each room, keeping personal and collaborative spaces separate.

3. User Authentication

- **Secure Sign-in/Sign-up:** Standard email/password authentication flow with OTP verification for password resets.
- **Google OAuth 2.0:** One-click sign-in and sign-up using Google for a frictionless user experience.
- **Session Management:** Uses secure, HTTP-only cookies with JWT for persistent sessions.

4. Modern, Responsive UI

- **Clean Interface:** Built with shadcn/ui, providing a modern, aesthetically pleasing, and responsive user experience.
- **Real-Time Feedback:** Uses Sonner for non-intrusive toast notifications for actions like "Copied!", "Logged in", etc.
- **Loading States:** Skeletons and loading spinners provide clear feedback during network requests.

Tech Stack

- **Frontend:**
 - **Framework:** React (with Vite)
 - **Language:** TypeScript
 - **State Management:** Redux Toolkit, RTK Query
 - **Real-Time:** Socket.IO Client
 - **Styling:** Tailwind CSS
 - **UI Components:** shadcn/ui
 - **Routing:** React Router
- **Backend:**
 - **Framework:** Express.js
 - **Language:** TypeScript (with ts-node)

- Real-Time: Socket.IO
- Database: MongoDB (with Mongoose)
- Authentication: JWT (JSON Web Tokens),
Google OAuth 2.0
- Deployment:
 - Frontend: Netlify
 - Backend: Render

GitHub Structure & Workflow

The project is structured as a monorepo with separate client and server directories.

```

/AirClip
├── client/
│   ├── src/
│   │   ├── components/
│   │   ├── features/ (Redux Slices)
│   │   ├── hooks/
│   │   ├── middlewares/ (SocketMiddleware)
│   │   └── ...
│   └── package.json
├── server/
│   ├── src/
│   │   ├── controllers/
│   │   ├── models/
│   │   ├── middleware/
│   │   └── index.ts
│   └── package.json
└── README.md
  
```

Frontend Repositories:

- **AirClip** (React Web App): Web-based application.
- **AirClip_react-native** (Mobile App): iOS and Android native application.
- **AirClip_react-electron** (Desktop App): Windows, macOS, and Linux desktop application.

Backend Repository:

- **AirClipServer**: Express.js backend server.

Branching Workflow:

- main: Stable, production-ready code.
- feature/...: For developing new features (e.g., feature/file-sharing).
- bugfix/...: For fixing bugs in the main branch.

Future Extensions

- File Sharing: Extend clipboard functionality to support sharing of small files and images.
- End-to-End Encryption: Implement E2EE for custom rooms to enhance privacy for sensitive collaboration.
- Persistent History: Save clipboard history to the database for each user and room.
- Progressive Web App (PWA): Wrap the application as a PWA for better mobile integration and offline access.

Contribution

AirClip is an open-source project licensed under the **AirClip Source License © 2025**.

GitHub Frontend: ([@vanshwebdev](#)) (Web app)

GitHub Backend: ([@vanshwebdev](#)) (Server)

GitHub react-native: ([@vanshwebdev](#)) (native Application)

GitHub react-electron: ([@vanshwebdev](#)) (Desktop app)

Contact

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